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ASDvit: Enhancing autism spectrum disorder classification using vision transformer models based on static features of facial images

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ABSTRACT

This study embarks on an exploratory journey into autism spectrum disorder (ASD), a multifaceted neurological developmental disorder with a spectrum of manifestations. Recognizing the transformative impact of early diagnosis and tailored medical interventions on the lives of children diagnosed with ASD and their families, The intersection of early diagnosis and tailored medical intervention can substantially enhance the quality of life for children diagnosed with ASD and their families. This study embarks on an innovative approach to augmenting the diagnostic process, specifically through the analysis of static features extracted from facial photographs of autistic children. By employing Vision Transformers (ViT) enhanced with Squeeze-and-Excitation (SE) blocks, our research delves into the potential of facial features as a biomarker for distinguishing autistic children from their typically developing counterparts. The fusion of ViT with SE mechanisms aims to amplify the model's sensitivity toward the subtle yet diagnostically crucial facial cues associated with ASD. Through comprehensive experimentation on a curated dataset, categorized into "autistic" and "non-autistic" groups, our approach demonstrates remarkable proficiency in identifying ASD, thereby opening new avenues for employing facial image analysis as a scalable biomarker in ASD diagnosis.

1. Introduction

Recently, artificial intelligence has significantly revolutionized early medical diagnosis and social health outcomes. By taking advantage of a large volume of medical data, these technologies enable the development of complex algorithms that can detect patterns and abnormalities related to the disease more accurately or earlier and help in better treatment of the disease. For instance, deep learning-based models excel in imaging diagnostics, such as early-stage cancer detection from radiographs and MRIs with remarkable accuracy. Machine learning algorithms can analyze patient history and genetic information to predict the likelihood of diseases, facilitating preventive care and personalized treatment plans. AI-powered systems also assist in streamlining administrative tasks, reducing errors, and improving efficiency in healthcare delivery. This integration of advanced technology not only enhances diagnostic precision and treatment effectiveness but also democratizes access to healthcare, ultimately leading to improved patient outcomes and a healthier society.

Artificial intelligence has made significant strides in the early diagnosis of autism spectrum disorder (ASD), providing invaluable support to patients and their families. Traditional diagnostic methods for autism often rely on lengthy observations and assessments, which can delay intervention. AI-driven tools, however, can analyze vast amounts of behavioral data, such as eye movement, facial expressions, and social interactions, to detect subtle signs of autism at an early age. Machine learning algorithms trained on diverse datasets can identify patterns indicative of ASD with high accuracy and enable earlier diagnosis and timely intervention. Early detection facilitated by AI enables the implementation of personalized therapeutic strategies, which can improve developmental outcomes for children with autism. Moreover, AI helps families manage these conditions by providing resources, support networks, and guidance, thereby reducing the emotional and logistical burden. Through these advancements, AI will not only enhance the quality of life for people with autism but also empower their families with the tools and knowledge to better support their loved ones. Artificial intelligence can provide faster, more accurate, and more objective assessments compared to traditional methods for early medical diagnosis of autism spectrum disorder (ASD). Traditional autism diagnosis often involves lengthy and subjective evaluations by specialists, which can lead to delays in identification and intervention. On the other

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hand, AI can analyze vast amounts of data from various sources such as genetic information, medical history, and behavioral patterns, with unprecedented speed and precision. Machine learning algorithms can identify subtle, early indicators of autism that may be overlooked by human observers, enabling earlier and more reliable diagnoses. Additionally, AI-driven tools can standardize diagnostic criteria, reducing variability and potential biases in human assessments. By streamlining the diagnostic process, AI not only expedites access to clinical early interventions but also alleviates the burden on healthcare professionals, allowing them to focus on personalized care and treatment planning. This technological advancement enhances the overall efficiency and effectiveness of autism diagnosis, ultimately leading to better outcomes for patients and their families.

2. Related work

During the last decade, the adoption of machine learning for computer-assisted diagnosis (CAD) systems has seen a marked increase in both clinical and scientific arenas. The application of neural networks, alongside advancements in transfer learning, has been pivotal in distilling dense, dimensional features from extensive datasets for further refinement of medical models. The technologies showcase the efficacy of neural networks in preventative measures against mental disorders [1]. Yet, the expansion of these neural network techniques into the realm of analyzing autism spectrum disorder (ASD) has faced hurdles, primarily due to ASD's intrinsic diversity and its complex cognitive manifestations [2].

Considerable focus has been placed on developing reliable indicators for ASD diagnosis, drawing significant attention in recent literature [2, 3]. The majority of neural network-driven ASD diagnoses have relied on neuroimaging techniques [4]. However, acquiring neuroimaging data frequently encounters obstacles, especially with the lack of cooperation from kids [5]. In response, recent innovations have pivoted towards behavioral analysis and affective computing to mitigate some of these challenges. These methods bank on the premise that individuals with ASD process attention and emotions differently when interacting with visual stimuli [3]. Specifically, in ASD-affected children, these variances are observable in facial expressions and eye movements [6], offering a less invasive alternative to neuroimaging by directly scrutinizing the patient's facial and eye-tracking data alongside behavioral cues. This approach not only aligns with but potentially enhances, established diagnostic frameworks like the Autism Diagnostic Observation Schedule (ADOS).

Neural networks have demonstrated their potential in differentiating ASD from non-ASD children, provided there is access to adequately labeled facial imagery [7]. Xie et al. [8] and Han et al. [9]employed neural network models utilizing eye-tracking data to forecast ASD, indicating the promising direction of in vitro image data (facial and eye-tracking) in crafting CAD systems for ASD. Radha et al. introduced a novel approach for behavioral autism detection, employing eye movement analysis techniques [10]. Gaspar et al. proposed a methodology centered on classifying Autism Spectrum Disorder (ASD) utilizing a kernel extreme learning machine (KELM) [11]. Their methodology included data augmentation to enhance training and further optimization of the KELM model using the Giza pyramids construction (GPC) method, leading to improved accuracy in classification. In a similar vein, Kollias et al. presented a transfer learning method specifically tailored for detecting ASD in high-functioning adults [12]. Their approach combined decision trees (DT), transfer learning techniques, and logistic regression on a dataset containing high-functioning ASD adults and control subjects. These studies represent diverse and innovative strategies aimed at enhancing early autism diagnosis and improving classification accuracy in different demographic groups.

Exploring the potential of facial characteristics as indicators for autism spectrum disorder (ASD) represents a cutting-edge direction in autism research. This approach hinges on the precise measurement of facial landmarks and the analysis of facial asymmetry through the distances between specific points on the face, aiming to establish a correlation with ASD through dysmorphology scores.

Miles and Hillman [13] introduced a novel classification system that categorizes children into "complex" or "essential" autism based on the presence of minor physical anomalies (MPAs). Children presenting with six or more MPAs, potentially with microcephaly, fall into the "complex" category, indicating a higher likelihood of genetic conditions, brain structural differences, seizures, and lower IQ levels. Conversely, those without signs of early embryological deviations were considered to have "essential" autism [14]. The researchers further developed the autism dysmorphology measure (ADM) to facilitate the identification of "complex" autism through a simplified process that does not necessitate direct physical examination but rather relies on photographic analysis [15].

In a related study, Angkustsiri, K. et al. [16] delved into the use of facial photography to identify generalized dysmorphology among a cohort of 324 children aged 2–5 years, encompassing those with ASD, developmental delays, and typically developing (TD) peers. By conducting physical assessments and setting a dysmorphology threshold based on the count of MPAs, they were able to segregate the children into dysmorphic and non-dysmorphic groups. Their findings underscored a significant prevalence of dysmorphic features among children with ASD compared to their TD counterparts ($p = 0.007$), suggesting that photographic analysis could serve as a tool for ASD symptom detection.

Building on this foundation, Tania et al. [17] embarked on pioneering work with the publication of "A transfer-learning-based autism classification model utilizing 2D face photos". The study leveraged a machine learning (ML) classifier to differentiate ASD from TD profiles, utilizing a comprehensive dataset from Kaggle comprising 2936 facial images of 1468 children diagnosed with ASD and an equal number of TD children. Through rigorous testing across multiple ML and deep learning classifiers, the MobileNet-V1 emerged as the top performer, achieving an accuracy of 90.67 %. Hosseini and colleagues assessed various convolutional neural networks (CNNs) for facial analysis in ASD-afflicted children, identifying models like MobileNet that achieved up to 90 % accuracy in distinguishing ASD from non-ASD individuals [18].

El Mouatasim and Ikermane [19] introduce approach presents a novel method for optimizing deep convolutional neural networks (CNNs) in autism facial detection using a Control Sub gradient Algorithm (CSA). The approach addresses challenges in training non-smoothed neural networks caused by activation functions like ReLU, where traditional optimizers may struggle. CSA dynamically adapts the learning rate during training by adjusting it upward or downward based on validation performance, ensuring faster convergence and improved stability. The leverage transfer learning with a pre-trained DenseNet-121 model, fine-tuning it on a limited dataset of facial images labeled for autism spectrum disorder (ASD). To enhance generalization, they incorporate regularization techniques to reduce model complexity. Experiments compare CSA with standard optimizers.

The literature underscores the distinctiveness of facial features in children with ASD compared to their TD peers. In recent years, transformers have emerged as leading models for Natural Language Processing (NLP) tasks, due to their scalability and computational efficiency. These models can be trained with over a hundred billion parameters without experiencing performance saturation. Inspired by the success of transformers in NLP and recognizing the potential benefits of the self-attention mechanism for image classification tasks, researchers proposed adapting the transformer architecture, with minimal modifications, for image classification [20]. This proposal led to the development of Vision Transformers (ViT), an architecture that divides images into 2D patches and feeds this linear sequence of patches into the model. The architecture of Vision Transformers is designed to leverage the strengths of the transformer framework for visual data processing.

In comparison, Convolutional Neural Networks (CNN) has long been

a popular tool for processing large volumes of data in computer vision. CNNs are composed of multiple layers and have shown strong performance in tasks such as object detection, image segmentation, and even some NLP applications [21]. A typical CNN architecture starts with convolutional layers that apply filters to the input image, extract low-level features like colors and edges, and progress to high-level feature extraction. Pooling layers then reduce the spatial dimensions, preserving essential features, and fully connected layers perform the final classification. CNNs are well-established in this field due to their ability to capture effectively spatial hierarchies in images.

Given the rising interest in Vision Transformers for image recognition tasks and the proven success of CNNs in image classification, this work aims to review and compare the state of the art in Vision Transformers (ViT) and Convolutional Neural Networks (CNN) for image classification. Transformers offer significant advantages, such as modeling long-range dependencies, adapting to varying input sizes, and enabling parallel processing, which make them suitable for image-related tasks. Moreover, the capabilities of Vision Transformer (ViT) models in image recognition hold promising potential for enhancing ASD diagnosis. However, ViTs perform poorly compared to CNNs on smaller datasets, a limitation that propels us to explore new methods. Inspired by these findings, here, we introduced an advanced ViT model using Squeeze-and-Excitation (SE) blocks to use facial image features to diagnose ASD in children.

3. Materials and methods

3.1. Data description

The cornerstone of our investigation into the application of Vision Transformers (ViT) augmented with Squeeze-and-Excitation (SE) blocks for Autism Spectrum Disorder (ASD) diagnosis Dataset Compiled by G. Piosenka [22] consists of facial images of children classified into two groups: those diagnosed with ASD and typically developing (TD) children. To compile this dataset, He tapped into publicly accessible image repositories, ensuring a diverse collection that spans various demographics, including age, gender, ethnicity, and a range of ASD severity levels The dataset contains 2926 images with binary labels (non-autism and autism) of resolution 224x224 pixels. It is split into a training set of 2526 images, a validation set of 200 images, and a test set of 200 images [22], we put test image = 0.05.

Before deploying the dataset for training and validation purposes, a series of preprocessing steps were applied to ensure data quality and uniformity. applying techniques to reduce potential biases that could affect the learning process. Recognizing the importance of data augmentation in preventing overfitting and enhancing model robustness, we implemented a series of transformations, including data augmentation techniques, to artificially expand our dataset without compromising the integrity of the data.

Following preprocessing, the dataset was systematically divided into training and testing segments. This division strategy was carefully crafted to ensure a balanced representation of ASD and TD samples across each set, thereby facilitating a comprehensive and unbiased evaluation of the model's diagnostic capabilities. We use transfer learning with the Vision Transformer (ViT) model, specifically the variant pre-trained on ImageNet-21k [23], comprising 14 million images across 21,843 classes, at a resolution of 224x224. This approach harnesses the rich visual representations learned from a vast and diverse dataset such as ImageNet-21k, which covers a wide range of object categories and visual concepts. By leveraging the knowledge of the pre-trained ViT models, we can take advantage of its ability to capture complex visual features and patterns, which are crucial for tasks such as image classification, detection, and segmentation, including medical image analysis such as autism diagnosis. Using transfer learning with a pre-trained ViT model enhances model performance, accelerates convergence during training, and reduces the need for extensive labeled data, making it a valuable asset in various image-related applications.

3.2. Implementation information

The ASDvit model was implemented in Python 3.9 using PyTorch 2.0.1 and CUDA 11.7 for GPU acceleration The code utilizes several libraries as shown below Table 1.

Also we use hyperparameters of the proposed model as shown in Table 2.

3.3. Proposed method

Vision Transformers (ViTs) have significantly advanced image classification, achieving state-of-the-art results by leveraging self-attention mechanisms to capture spatial relationships within images. However, the standard ViT architecture primarily emphasizes spatial correlations between features within a single channel, often neglecting the critical interdependence among channels. This limitation can lead to suboptimal feature representation, particularly in tasks requiring nuanced analysis, such as Autism Spectrum Disorder (ASD) detection via facial features. To address this challenge, the ViT model has been innovatively enhanced with Squeeze-and-Excitation (SE) Blocks [24] introducing a channel-attention mechanism designed to refine the importance of feature channels dynamically. The SE Blocks perform feature recalibration through two key operations: (1) Squeeze, which aggregates spatial information by compressing feature maps into a compact channel descriptor, and (2) Excitation, which captures channel-wise dependencies by applying a fully connected layer to the squeezed output. This recalibration process allows the model to prioritize channels that contribute more significantly to the classification task, thus enhancing its sensitivity and specificity in identifying ASD-related facial features. In this improved architecture, each ViT encoder layer is augmented with an SE Block, positioned immediately after the multi-head self-attention mechanism. This integration ensures that both spatial and channel-wise relationships are effectively modeled, resulting in superior diagnostic performance and a more robust feature representation. The core functionality of SE Blocks lies in their ability to selectively emphasize informative channels while suppressing less relevant ones, offering a novel approach to enhancing ViT's capacity for complex image classification tasks. The core principle behind SE Blocks can be described by the following:

The squeeze operation involves global spatial pooling to generate channel-wise descriptors as shown in Fig. 1. Let X is the input feature map with dimensions $C \times H \times W$, where C is the number of channels, H is the height, and W is the width, The global average pooling (GAP) operates applied to X to compute the noise descriptor z: $= GAP(X) = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{w} X_{ij}$

Here, Xij represents the activation at position (i,j) in the feature map X.

The Excitation (Channel-Wise Recalibration) Operation consists Two fully connected (FC) layers process the squeezed vector: Bottleneck Layer: Reduces dimensionality by a reduction ratio, followed by ReLU

Table 1
Libraries used in the model.

Library	version
Torch	2.0.0
Transformers	4.30.0
Datasets	2.12.0
pandas	1.5.3
Numpy	1.23.5
scikit-learn	1.2.2
imbalanced-learn	0.10.1
Matplotlib	3.7.1
Evaluate	0.4.0
Accelerate	0.20.3

Table 2
Hyperparameters That used in training model.

Parameter	Value
Optimizer	AdamW
Learning Rate	5e-6
Weight Decay	0.02
Warmup Steps	50
Batch Size (Train)	32
Batch Size (Eval)	8
Evaluation Strategy	Per epoch
Save Strategy	Per epoch
Augmentation Techniques	Resize, Random Rotation, Adjust Sharpness, Normalize, To Tensor
Criterion (Loss Function)	Cross Entropy Loss
Learning Scheduler	Linear with warmup (default)

activation and Channel-Scaling Layer: Restores original dimensionality using a sigmoid-activated FC layer, generating per-channel scaling factors between 0 and 1. Let $fsq(z)$ denote the output of the squeeze operation (channel-wise descriptor). The excitation operation is defined as follows.

- Apply the first FC layer with a sigmoid activation function to generate the "excitation" vector s:

$$s = \sigma(W1 \cdot fsq(z) + b1)$$

where $W1$ and $b1$ are the weights and biases of the first FC layer, respectively, and σ represents the sigmoid activation function.

- Apply the second FC layer with a ReLU activation function to modulate the input feature map X using the excitation vector s to obtain the recalibrated feature map Y:

$$Y = \mathrm{ReLU}(W2 \cdot s + b2) \cdot X$$

where $W2$ and $b2$ are the weights and bias of the second FC layer, respectively, and ReLU represents the rectified linear unit activation function.

The final output Y represents the feature map with recalibrated channel-wise information, emphasizing important channels that contribute significantly to the network's decision-making process.

The Feature Recalibration The scaling factors are element-wise multiplied with the original self-attention output, amplifying critical channels and suppressing redundant ones. This recalibrated tensor is then passed to the encoder's feedforward network and the Hierarchical Feature Learning with SE-Enhanced Encoders, The ViT encoder comprises 12 identical layers, each equipped with an SE block. As the network deepens the lower layers use SE to prioritize local texture and edge features and the higher layers leverage SE to emphasize global anatomical structures indicative of ASD. For optimal performance, SE Blocks are strategically placed between the multi-head attention module and the normalization layer within each ViT encoder layer as shown in Fig. 2. This placement provides the following benefits: The SE Block operates on the output of the multi-head attention, which has already identified relevant features based on spatial relationships. This allows the SE Block to refine the importance of these features based on their channel-wise interactions, The SE Block analyzes the importance of these features across channels, highlighting informative channels and suppressing less relevant channels. This leads to a more discriminative feature representation, The normalization layer, following the SE Block, ensures a well-behaved distribution of the refined features for subsequent processing in the encoder layer. This helps maintain numerical stability during training and inference, by incorporating SE Blocks, ViTs can effectively capture both spatial and channel-wise relationships within the input image. The main key Innovations of ASDvit Integration is the dynamic Channel Attention: Unlike static self-attention weights, SE blocks adaptively recalibrate channels at multiple abstraction levels, allowing the model to progressively refine its focus on diagnostically relevant features and Complementarity with Self-Attention: While self-attention captures long-range patch dependencies, SE blocks specialize in modeling inter-channel relationships, creating a dual mechanism for feature optimization, This enhanced feature representation can potentially lead to improved classification accuracy compared to standard ViT

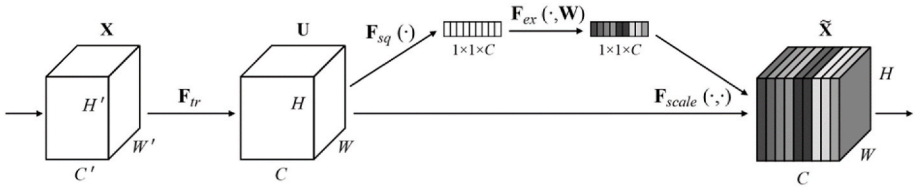

Fig. 1. Representation of a Squeeze-and-Excitation block.

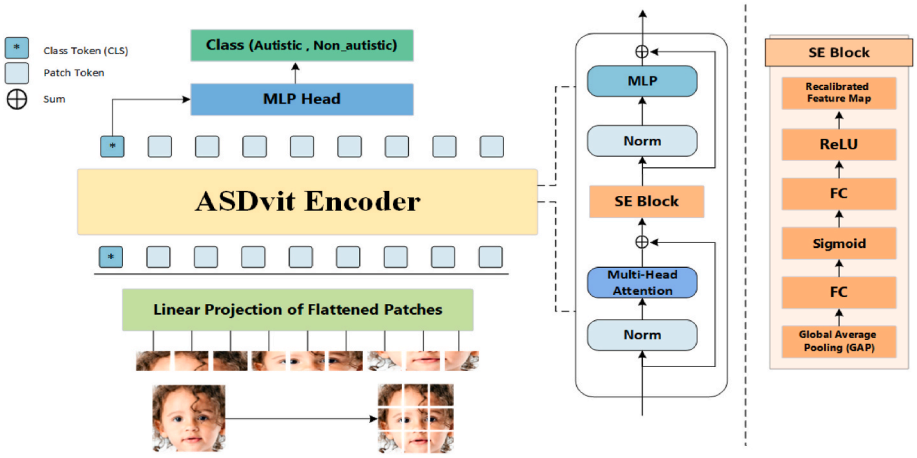

Fig. 2. The overall diagram of the proposed ASDvit model.

architectures.

4. Results and discussion

In Our study, we used the ASD dataset by G. Piosenka [22] which includes 2926 labeled images of children's faces, divided into ASD and typically developing groups. The dataset is divided into 2526 training images, 200 validation images, and 200 test images, we put test image = 0.05. The use of a pre-trained Vision Transformer can be viewed as a neutral point in our results, especially compared to other models, given that our model is also based on the ImageNet-21k dataset. This similarity in the training data across different models minimizes the potential differences attributed solely to the pre-training phase. Instead, the focus shifts to model architecture, hyperparameters tuning, and fine-tuning strategies that play a crucial role in determining performance outcomes. Therefore, while the pre-trained Vision Transformer provides a solid foundation for feature extraction and learning, the actual performance gain or loss is more influenced by specific implementation choices and optimizations within the model pipeline.

We embarked on an extensive evaluation of various architectures to ascertain their effectiveness in the automated diagnosis of Autism Spectrum Disorder (ASD) using facial images. Central to our investigation was the comparison of the Area Under the Receiver Operating Characteristic curve (AUC) as a primary metric for diagnostic performance because the illustrates the performance of a binary classification model across different classification thresholds [25]. In medical image classification tasks, AUC is preferred over accuracy, which is derived from the confusion matrix, because accuracy does not account for the prediction probability, and it assumes that the class with the highest probability estimation is the target class. Moreover, the accuracy of a model is influenced by the size of the test dataset. Research indicates that when the test dataset is small, AUC serves as a superior accuracy metric [26]This metric provides an insightful measure of a model's ability to distinguish between ASD and typically developing (TD) individuals, with a higher AUC indicating superior diagnostic accuracy, The same dataset was used in all models and also was used in our model to clarify the comparison results [22].

Among the models evaluated in Table 3 shows, the VGG-19 architecture, renowned for its depth and complexity with approximately 139 million parameters and pre-trained on the extensive ImageNet-21k dataset, achieved an impressive AUC of 93.65 [27,28]. Following closely, the ResNet50 model, with its comparatively leaner structure of 23.5 million parameters, demonstrated a slightly higher AUC of 94.82 when also pre-trained on ImageNet-21k [28]. The MobileNetV3 model, notable for its efficiency and reduced parameter count of 4.2 million, showcased an AUC of 94.43, underscoring the potential of lightweight models to achieve high diagnostic performance [7,27,29].

zFurther evaluations included the EfficientNet-B4 model, which, with 17.6 million parameters, attained an AUC of 95.13 when pre-trained on ImageNet-21k [30]. Not to be outdone, the Xception architecture, wielding 20.8 million parameters, reached a pinnacle AUC of 95.40, setting a high benchmark for subsequent models [27].

Our novel contribution to this field, the ASDvit model, integrates the ViT-B backbone with Squeeze-and-Excitation blocks, maintaining the parameter count of 85.8 million but uniquely pre-trained on ImageNet-21k. This model set a new benchmark in our study, achieving an unprecedented AUC of 97.77 as shown in Table 4. This remarkable performance not only highlights the advantage of incorporating attention recalibration mechanisms through SE blocks but also reaffirms the potential of large transformer models, especially when coupled with innovative architectural enhancements, in accurately diagnosing ASD from facial images.

Table 3
Comparison of ASDvit and other methods based on the (ASD) children's dataset.

Methods	Backbone	Params	Pretrained	AUC
[27,28]	VGG-19	139M	ImageNet-21k	93.65
[28]	ResNet50	23.5M	ImageNet-21k	94.82
[7,27,29]	MobileNetV3	4.2M	ImageNet-21k	94.43
[30]	EfficientNet-B4	17.6M	ImageNet-21k	95.13
[27]	Xception	20.8M	ImageNet-21k	95.40
Our ASDvit	Vit-B	85.8M	ImageNet-21k	**97.77**

Among the models evaluated, Xception consistently demonstrates strong performance across various metrics. The AUC value of 96.63 % confirms Xception's excellent overall performance in discriminating between the two classes, EfficientNetB1 also stands out with competitive performance, particularly in terms of AUC (95.06 %) [30], MobileNet, although exhibiting balanced, falls slightly behind Xception and EfficientNetB1 in terms of overall performance, as reflected in its lower AUC value (92.81 %), Training and validation accuracy both go up as the number of epochs increases. At first, the training accuracy rises until it reaches over 90 %. and the validation accuracy follows a similar trend. The training loss increases slowly, while the validation loss drops gradually, Fig. 3 shows that the general trend points to a well-trained model that does a good job.

Our model ASDvit introduces a novel integration of the Squeeze-and-Excitation (SE) method with the Vision Transformer (ViT) architecture. This integration aims to enhance the model's feature learning capabilities and improve its performance in classifying Autistic and Non-Autistic images.

The results are shown in Fig. 3. For a fair comparison and based on the same dataset, we used values for other methods reported from Ref. [23]. It demonstrates that our model achieves competitive performance compared to established models like Xception and EfficientNetB1. With a sensitivity of 87.67 % and specificity of 94.95 %, strikes a balance between accurately identifying Autistic cases while also effectively distinguishing Non-Autistic cases. This is further supported by its high AUC value of 97.77 %, indicating strong discriminatory power between the two classes.

Upon examining the prediction results as shown in Fig. 4, several key observations can be made. ASDvit demonstrates the highest result in predicting both Autistic and Non-Autistic cases, with 70 % for Autistic images and 64 % for Non-Autistic images. This indicates the model's robustness in correctly identifying individuals with ASD as well as those without ASD. To ensure a fair comparison, we utilized values from other methods reported using the same dataset from Ref. [23] (see Fig. 5).

EfficientB2, while achieving for Autistic images (66 %), shows a (42 %) for Non-Autistic images, Xception and EfficientNetB1 exhibit similar performance in predicting Autistic cases, with 55.33 % and 53.33 % respectively. However, Xception shows a (47.33 %) for Non-Autistic cases compared to EfficientNetB1 (57.33 %)., EfficientNetB0 achieves a relatively balanced accuracy for both Autistic (59.33 %) and Non-Autistic (58.67 %) cases, suggesting a good overall performance in prediction, MobileNet, while showing for Non-Autistic cases (58.67 %), lags in predicting Autistic cases (42 %). This large difference between the models indicates a difference in the diagnostic models for autism [30].

These results underscore the transformative potential of advanced neural network architectures, particularly Vision Transformers augmented with Squeeze-and-Excitation blocks, in refining the accuracy and reliability of ASD diagnosis through non-invasive facial image analysis. The ASDvit model, with its unparalleled AUC, represents a significant leap forward in the field, promising to enhance early diagnostic practices and contribute to the broader goal of accessible and efficient healthcare solutions for ASD.

4.1. Discussion

The advancements in computer-assisted diagnosis (CAD) systems, particularly through the lens of deep learning technologies, have opened new vistas in the medical field, offering promising avenues for the early

Table 4
Comparison of the proposed method with other methods based on the G.Piosenka ASD children dataset [22].

Methods	Backbone	sensitivity	specificity	NPV	PPV	AUC
[30]	MobileNet	86.11	83.33	86.66	82.66	92.81
	EfficientNetB0	84.71	88.11	84	88.66	93.38
	EfficientB2	84.93	93.28	83.33	94	94.31
	EfficientNetB1	86.06	94.07	84.66	94.66	95.06
	Xception	88.46	91.66	88	92	96.63
Our ASDvit		**87.67**	**94.95**	**88.6**	**94.11**	**97.77**

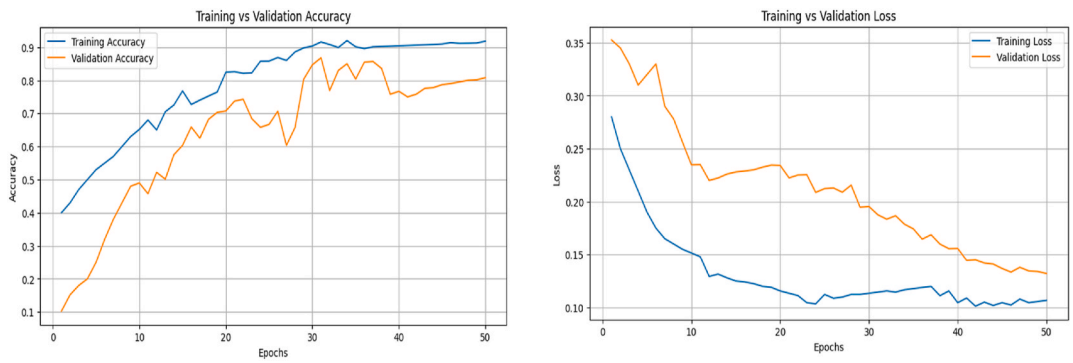

Fig. 3. Accuracy and loss values for ASDvit model.

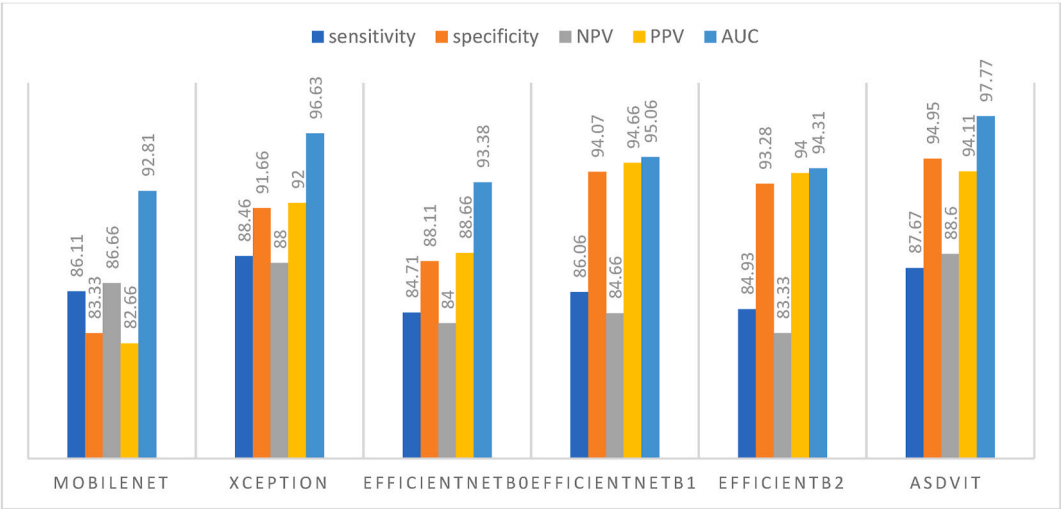

Fig. 4. Comparison of the proposed methods with other methods.

detection and diagnosis of Autism Spectrum Disorder (ASD). This study's exploration into various machine learning architectures, and the development of the ASDvit model, represents a significant leap forward in leveraging artificial intelligence for healthcare applications. Using transfer learning with the Vision Transformer (ViT) model, specifically the Google vit variant pre-trained on ImageNet-21k, offers several significant advantages. First, leveraging a pre-trained ViT model allows us to capitalize on extensive visual knowledge and representations learned from a large-scale dataset such as ImageNet-21k, which comprises 14 million images across 21,843 classes. This pre-training process enables the ViT model to capture a wide range of visual features and patterns, making it adept at handling diverse image-related tasks. Furthermore, transfer learning with a pre-trained ViT model accelerates the model's convergence during training on specific tasks, such as image classification or segmentation for medical applications like autism detection. This fast convergence not only reduces training time and computational resources but also increases the performance and generalization capabilities of the model, leading to more accurate and reliable results. Overall, the integration of transfer learning with a pre-trained ViT model significantly contributes to the efficiency and effectiveness of image analysis tasks and shows its value in various domains including medical imaging and diagnostics.

The comparative analysis of traditional convolutional neural networks (CNNs) such as VGG-19, ResNet50, MobileNetV3, EfficientNet-B4, and Xception against the newer Vision Transformer (ViT) models underscores a pivotal shift in the approach to medical image analysis. While CNNs have long been the backbone of image classification tasks, their inherent limitations, particularly in capturing global contextual information, have prompted researchers to explore alternative architectures. The ResNet50 [28]and VGG-19 models [27,28], with their deep layers and extensive parameters, demonstrated robust performance, as evidenced by their AUC scores of 94.82 and 93.65, respectively. However, the efficiency and scalability of these models in clinical settings are constrained by their computational demands and the need

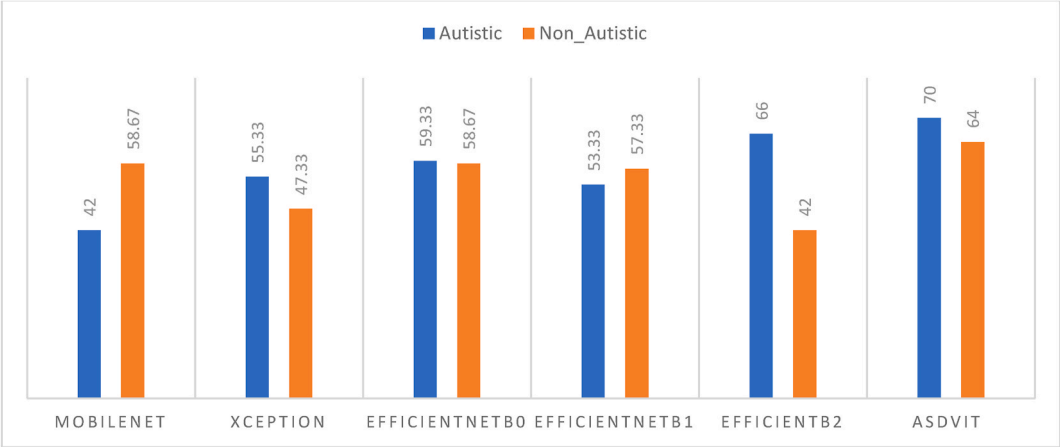

Fig. 5. Comparison of the proposed method with other models in terms of accuracy for separate classes of Autistic and Non-Autistic.

for substantial data augmentation.

In contrast, the MobileNetV3 [7,27,29] and EfficientNet-B4 [30]architectures presented a balance between performance and computational efficiency, with AUC scores of 94.43 and 95.13, respectively, highlighting the potential for deploying lightweight models in resource-constrained environments. Nevertheless.

The models considered in this study are MobileNet, Xception, EfficientNetB0, EfficientNetB1, EfficientB2 [30], and a Our model denoted as ASDvit, The performance metrics, including sensitivity, specificity, negative predictive value (NPV), positive predictive value (PPV), and area under the ROC curve (AUC), provide insights into the models' ability to accurately classify images. Sensitivity measures the model's ability to correctly identify true positive cases (Autistic), while specificity measures its ability to correctly identify true negative cases (Non-Autistic). NPV and PPV indicate the model's ability to correctly predict negative and positive cases, respectively.

Among these models, Xception emerges as a standout performer, achieving high sensitivity (88.46 %) and specificity (91.66 %). This indicates its robust ability to correctly identify both Autistic and Non-Autistic cases. Additionally, Xception exhibits a balanced negative predictive value (NPV) of 88 % and positive predictive value (PPV) of 92 %, showcasing its reliability in predicting negative and positive cases accurately. The area under the ROC curve (AUC) for Xception is notably high at 96.63 %, reaffirming its overall excellent performance in discriminating between the two classes, EfficientNetB1 also demonstrates competitive performance, particularly excelling in specificity (94.07 %) and PPV (94.66 %). This implies its high accuracy in correctly identifying Non-Autistic cases. However, its sensitivity is slightly lower compared to Xception, indicating a marginally higher rate of false negatives, MobileNet, while showcasing balanced sensitivity and specificity, falls marginally behind Xception and EfficientNetB1 in overall performance, as indicated by its lower AUC value of 92.81 %, EfficientB2 exhibits commendable specificity (93.28 %) but lower sensitivity (84.93 %), suggesting a higher likelihood of missed Autistic cases (false negatives) compared to other models.

ASDvit the custom model integrating the Squeeze-and-Excitation (SE) method with the Vision Transformer (ViT) architecture, demonstrates promising results with high sensitivity (87.67 %) and specificity (94.95 %). This indicates its effectiveness in accurately identifying both Autistic and Non-Autistic cases. Moreover, ASDvit showcases a balanced NPV (88.6 %) and PPV (94.11 %), highlighting its reliability in predicting negative and positive cases accurately. The AUC value for ASDvit is notably high at 97.77 %, indicating its exceptional discriminative power in distinguishing between Autistic and Non-Autistic images.

This model's exceptional performance can be attributed to several key factors. Firstly, the SE blocks enhance the model's focus on salient features within the facial images by recalibrating channel-wise feature responses, thereby improving the sensitivity and specificity of ASD detection. Secondly, the large parameter space of the ViT-B backbone, coupled with pretraining on the ImageNet-21k dataset, provides a rich feature representation that is finely tuned for the task at hand, The implications of this study extend far beyond the technical achievements. The ASDvit model's high diagnostic accuracy opens up new possibilities for non-invasive, efficient, and scalable screening tools for ASD. Early detection of ASD is critical for timely intervention, potentially altering developmental trajectories and improving outcomes for children with ASD. By providing a reliable, automated diagnostic tool, this model can supplement traditional diagnostic methods, reducing the reliance on scarce specialized personnel and enabling more widespread screening efforts, particularly in underserved regions, Looking forward, the scope for employing Vision Transformers in ASD diagnosis extends beyond facial analysis. Integrating multimodal data, including genetic, neuroimaging, and behavioral assessments, can provide a more holistic view of ASD, potentially unveiling novel biomarkers and improving diagnostic precision. Moreover, the adaptive nature of ViT models presents opportunities for personalized medicine approaches, tailoring diagnostic and intervention strategies to individual profiles.

5. Conclusions

The early diagnosis of autism spectrum disorder can substantially enhance the quality of life for children diagnosed with ASD and their families. This research introduces an innovative approach to augmenting the diagnostic process, specifically through the analysis of static features extracted from facial photographs of autistic children. The proposed method ASDvit employs Vision Transformer models to enhance the early diagnosis of ASD and, utilizing transfer learning with a pre-trained ImageNet-21k offers significant benefits. This approach capitalizes on the model's extensive visual knowledge, leading to faster training convergence and improved performance in tasks such as autism detection. By leveraging pre-trained representations, the ViT proposed model demonstrates enhanced efficiency, accuracy, and generalization, making it a valuable asset in medical imaging and analysis. By harnessing the power of advanced machine learning techniques, we can pave the way for more accessible, accurate, and efficient diagnostic tools, ultimately contributing to better care and outcomes for individuals with ASD. As we continue to explore the frontiers of artificial intelligence in healthcare, the integration of technological innovations with clinical expertise will undoubtedly play a pivotal role in shaping the future of medical diagnostics.

CRediT authorship contribution statement

Hayder Ibadi: Writing – original draft, Software, Methodology. **Amir Lakizadeh:** Supervision, Conceptualization.

Ethical Statement

Hereby, we insert author name/consciously assure that for the manuscript/insert title/the following is fulfilled:

1) This material is the authors' own original work, which has not been previously published elsewhere.
2) The paper is not currently being considered for publication elsewhere.
3) The paper reflects the authors' own research and analysis in a truthful and complete manner.
4) The paper properly credits the meaningful contributions of co-authors and co-researchers.
5) The results are appropriately placed in the context of prior and existing research.
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we agree with the above statements and declare that this submission follows the policies of the Journal Ethics as outlined in the Guide for Authors and in the Ethical Statement.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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